BRAC University

Department of Mathematics and Natural Sciences

LECTURE ON

Real Analysis II (MAT321)

Theorems on Uniform Convergence

Continuity, Integration, Differentiation

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CONDUCTED BY

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Convergence of Sequence in Real Numbers

Convergence of a Sequence

A sequence $\{a_n\}$ is said to **converge** to a real number a if for every real number $\varepsilon > 0$, there exists a natural number $N \in \mathbb{N}$ such that for all $n \geq N$, the terms of the sequence satisfy the inequality

$$|a_n - a| < \varepsilon.$$

If such a number a exists, it is called the **limit** of the sequence, and we write

$$\lim a_n = a \quad \text{or} \quad a_n \to a \text{ as } n \to \infty.$$

Cauchy Criterion for Convergence

A sequence $\{a_n\}$ is said to satisfy the **Cauchy criterion** if for every real number $\varepsilon > 0$, there exists a natural number $N \in \mathbb{N}$ such that for all $m, n \geq N$, the terms of the sequence satisfy the inequality

$$|a_n - a_m| < \varepsilon.$$

Note: A sequence that satisfies the Cauchy criterion is called a **Cauchy sequence**. In the real numbers, every Cauchy sequence converges to a limit, and every convergent sequence is a Cauchy sequence.

Limit and Continuity of a function

Limit of a Function

Let $f : A \to \mathbb{R}$ be a function defined on a subset A of the real numbers, and let c be a limit point of A. We say that the **limit** of $f(x)$ as x approaches c is equal to L, and we write

$$\lim_{x \to c} f(x) = L,$$

if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that for all $x \in A$ with $0 < |x - c| < \delta$, we have

$$|f(x) - L| < \varepsilon.$$

Continuity of a Function

A function $f : A \to \mathbb{R}$ is said to be **continuous** at a point $c \in A$ if

$$\lim_{x \to c} f(x) = f(c).$$

In other words, f is continuous at c if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that for all $x \in A$ with $|x - c| < \delta$, we have

$$|f(x) - f(c)| < \varepsilon.$$

Poinwise Convergence and Uniform Convergence

Pointwise Convergence

A sequence of functions $\{f_n\}$ defined on a domain D is said to **converge pointwise** to a function f on D if for **every** $x \in D$ and for every $\varepsilon > 0$, there exists a natural number $N \in \mathbb{N}$ (which may depend on x and ε) such that for all $n \geq N$, the following inequality holds:

$$|f_n(x) - f(x)| < \varepsilon.$$

In this case, we write

$$\lim_{n \to \infty} f_n(x) = f(x) \quad \text{for all } x \in D.$$

Uniform Convergence

A sequence of functions $\{f_n\}$ defined on a domain D is said to **converge uniformly** to a function f on D if for every $\varepsilon > 0$, there exists a natural number $N \in \mathbb{N}$ such that for all $n \geq N$ and for all $x \in D$, the following inequality holds:

$$|f_n(x) - f(x)| < \varepsilon.$$

In this case, we write

$$f_n \to f \text{ uniformly on } D.$$

Note: In uniform convergence, the choice of N depends only on ε and not on x, which is a stronger condition than pointwise convergence.

Uniform Convergence implies Pointwise Convergence

Theorem

If a sequence of functions $\{f_n\}$ converges uniformly to a function f on a domain D, then it also converges pointwise to f on D.

Uniform Limit of Continuous Functions is Continuous

Theorem

Let $\{f_n\}$ be a sequence of functions defined on a domain D, and suppose that each f_n is continuous at a point $c \in D$. If $\{f_n\}$ converges uniformly to a function f on D, then f is also continuous at the point c.

Problem

Show that $f_n(x) = x^n$ is not uniformly convergent on $[0, 1]$.

Interchange of Limits under Uniform Convergence

Interchange of Limits for Sequence

Let $\{f_n\}$ be a sequence of functions that converges uniformly to a function f on a set $D \subset \mathbb{R}$, and let x_0 be a limit point of D. Suppose that for each $n = 1, 2, 3, \ldots,$

$$\lim_{x \to x_0} f_n(x) = a_n$$

exists. Then:

1. the sequence $\{a_n\}$ converges, and
2. $$\lim_{x \to x_0} f(x) = \lim_{n \to \infty} a_n.$$

Interchange of Limits for Series

Let $D \subset \mathbb{R}$ and let $\sum_{n=1}^{\infty} f_n$ be a series of functions defined on D whose partial sums

$$S_N(x) = \sum_{n=1}^{N} f_n(x)$$

converge uniformly on D to a function $f : D \to \mathbb{R}$. Let x_0 be a limit point of D. Assume that for each $n = 1, 2, 3, \ldots$ the limit

$$\lim_{\substack{x \to x_0 \\ x \in D}} f_n(x) = a_n$$

exists. Then:

1. the numerical series $\sum\limits_{n=1}^{\infty} a_n$ converges, and
2. $$\lim_{\substack{x \to x_0 \\ x \in D}} f(x) = \sum_{n=1}^{\infty} a_n.$$

When Pointwise Convergence is Uniform Convergence

Dini's Theorem

Let K be compact, and $f_n : K \to \mathbb{R}$ continuous. Suppose $f_n \to f$ pointwise on K, f is continuous, and the convergence is monotone (either $f_n \uparrow f$ or $f_n \downarrow f$ pointwise). Then $f_n \to f$ uniformly on K.

Problem

Let $x \geq 0$ and define $f_1(x) = \sqrt{x}$, $f_{n+1}(x) = \sqrt{x + f_n(x)}$. Discuss uniform convergence on $[0, 1]$.

Uniform Convergence and Integration

Uniform Convergence and Integration

Let $\{f_n\}$ be a sequence of functions that converges uniformly to f on $[a, b]$. If each function f_n is integrable on $[a, b]$, then f is integrable on $[a, b]$, and the sequence

$$\left\{\int_a^x f_n(t)\,dt\right\}$$

converges uniformly to

$$\int_a^x f(t)\,dt$$

on $[a, b]$, that is,

$$\int_a^x f(t)\,dt = \lim_{n\to\infty} \int_a^x f_n(t)\,dt, \qquad \forall x \in [a, b].$$

Uniform Convergence of a Series and Integration

If a series $\sum_{n=1}^{\infty} f_n$ converges uniformly to f on $[a, b]$, and each term $f_n(x)$ is integrable on $[a, b]$, then f is integrable on $[a, b]$, and the series

$$\sum_{n=1}^{\infty} \left(\int_a^x f_n(t)\, dt \right)$$

converges uniformly to

$$\int_a^x f(t)\, dt$$

on $[a, b]$, that is,

$$\int_a^x f(t)\, dt = \sum_{n=1}^{\infty} \left(\int_a^x f_n(t)\, dt \right), \qquad \forall x \in [a, b].$$

Problem

Define on $[0,1]$:

$$f_n(x) = \begin{cases} n(1-nx), & 0 < x < \frac{1}{n}, \\ 0, & \text{otherwise.} \end{cases}$$

Compute

$$\int_0^1 f_n(x)\,dx$$

and compare it with the pointwise limit of $\{f_n\}$.

Problem

Let

$$f_n(x) = nxe^{-nx^2} \quad \text{on } [0, 1].$$

Compute

$$\int_0^1 f_n(x)\,dx$$

and compare it with the pointwise limit of $\{f_n\}$.

Uniform Convergence and Differentiation

Uniform Convergence of Derivatives

Let $\{f_n\}$ be a sequence of functions defined on the closed interval $[a, b]$ such that $f_n \to f$ pointwise on $[a, b]$, and assume that each f_n is differentiable. If $\{f_n'\}$ converges uniformly on $[a, b]$ to a function G, then f is differentiable on $[a, b]$ and

$$f'(x) = G(x).$$

Uniform Convergence of Derivatives with One-Point

Let $\{f_n\}$ be a sequence of differentiable functions on $[a,b]$ such that it converges at least at one point $x_0 \in [a,b]$. If the sequence of derivatives $\{f_n'\}$ converges uniformly to a function G on $[a,b]$, then the sequence $\{f_n\}$ converges uniformly on $[a,b]$ to a function f and

$$f'(x) = G(x).$$

Term-by-Term Differentiation of a Series

Let $\sum_{n=1}^{\infty} f_n$ be a series of differentiable functions on $[a, b]$ such that it converges at least at one point $x_0 \in [a, b]$. If the series of derivatives

$$\sum_{n=1}^{\infty} f_n'$$

converges uniformly to a function G on $[a, b]$, then the given series

$$\sum_{n=1}^{\infty} f_n$$

converges uniformly on $[a, b]$ to a function f and

$$f'(x) = G(x).$$

Problem

Let

$$f_n(x) = \frac{\cos(nx)}{n} \qquad (x \in \mathbb{R}).$$

1. Show that $f_n \to 0$ uniformly on $\mathbb{R}$.
2. Find where f_n' converges.
3. Construct a modified example where $f_n \to 0$ uniformly but $\{f_n'\}$ is unbounded.

Uniform Convergence and Differentiation

State the theorem that relates uniform convergence and differentiation for a sequence of functions. Hence verify the result in detail for the sequence of functions $\{f_n\}$ where

$$f_n(x) = x - \frac{x^n}{n}, \qquad x \in [0, 1].$$

Thank You!

We'd love your questions and feedback.

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References

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